Agile Team Charter

SAEON Rainfall Monitoring Project

1. Team Purpose & Vision

Why do we exist as a team?

- **Team Mission:** To develop an ESP32 rainfall monitoring prototype for SAEON, publishing remote sensor data through cellular MQTT to a live dashboard.
- **Problem We Are Solving:** Manual retrieval from USB data loggers delays rainfall information and requires visits to remote SAEON sites.
- **Target Users:** SAEON researchers and environmental scientists; beneficiaries include government agencies, conservation organisations, and local communities.

Success Definition

- A working prototype counts rain-gauge tips and displays live data.
- Positive feedback from stakeholders and lecturer.
- All project deliverables completed on time.
- Remote monitoring reduces reliance on manual data retrieval.

Product / Service Definition

SAEON Real-Time Rainfall Monitoring System including:

- **Hardware: ESP32 WROOM-32, TR-525M tipping bucket rain gauge, DS18B20 temperature sensor and SIM7000G cellular module. LilyGO was an earlier controller option.**
- **Core Capabilities: Rainfall and temperature sensing, cellular MQTT through HiveMQ Cloud, Supabase data storage, a live dashboard, historical records, CSV export, authenticated access and station-health information.**
- **Testing Setup: Breadboard, jumper wires, USB power and reference measurements. DHT11 and other available temperature/humidity sensors were exploratory test options; DS18B20 is the selected temperature sensor.**
- **Deployment Scope: Final enclosure and installation, baseline field calibration and 48-hour field validation remain In Progress. Wind, UV, SHT31 humidity and BMP280/BME280 pressure sensing are future enhancements.**
- **Outdoor Provisions: Battery, charging, solar integration, voltage regulation and enclosure preparation are included in the current scope. Final outdoor acceptance depends on site access, power and protection. GPS and over-the-air updates remain future enhancements.**

2. Team Members & Roles

Name	Role	Key Responsibilities
Redah Gamieldien	Projects Leader	Oversees project progress, manages requirements, ensures goals are met
Angelo Adams	Frontend Developer / Designer	UI/UX design, frontend system development, wireframes
Qaasim Isaacs	Backend Developer / Tester	Backend system development, testing, quality assurance
Ethan Williams	Analyst / Researcher / Writer	Research, data gathering, validation of problem and solution, documentation
Rameez Karriem	Agile Lead / Presenter	Facilitates meetings, ensures Agile practices, presents deliverables

3. Ways of Working

3.1 Agile Framework

- **Framework: Kanban task tracking in Trello with weekly sprint planning. GitHub stores source code, documentation and evidence.**

3.2 Ceremonies

Ceremony	Frequency	Purpose
Daily Stand-up	3 times per week	Share progress, next steps
Planning	Weekly	Plan tasks and assign responsibilities
Review / Demo	End of each milestone	Present progress and get feedback
Retrospective	After each milestone	Reflect and improve team performance

4. Working Agreements

- Respect deadlines and commitments.
- Communicate delays early.
- Participate actively in meetings.
- Complete assigned tasks on time.
- Support teammates when needed.
- Maintain quality in all deliverables.

5. Communication

- **Primary Channels:** In-person meetings and WhatsApp messaging.
- **Response-Time Expectations:** Within 20 to 30 minutes depending on urgency.
- **Remote / Hybrid Norms:** Meet on Microsoft Teams if unable to meet in person.

6. Decision-Making

- Decisions are made through team discussion and majority vote.
- Product Owner (Group Leader) makes final decision if no agreement is reached.
- All team members are encouraged to contribute ideas.

7. Definition of Done (DoD)

- Code / design is completed.
- Work is reviewed by team members.
- Testing is completed successfully.
- Documentation is updated.
- Rain-gauge counting, MQTT transmission, and dashboard requirements are met.
- Approved by Project Leader.
- Ready for presentation / submission.

8. Core Values & Behaviours

Core Values

- **Respect:** Value each team member's input.
- **Transparency:** Be open about progress and challenges.
- **Accountability:** Take responsibility for assigned tasks.
- **Trust:** Support and rely on each other.
- **Learning Mindset:** Continuously improve and adapt.

Agile Principles

- Deliver working solutions incrementally.

- Embrace change and adapt quickly.
- Maintain continuous communication.
- Focus on user needs and value delivery.
- Reflect and improve regularly (retrospectives).

9. Goals and Milestones

SMART Goals

- **Specific:** Develop an ESP32 rainfall monitoring prototype for SAEON.
- **Measurable:** Demonstrate tip counting, MQTT delivery, and live dashboard updates.
- **Achievable:** Assign tasks based on team skills and availability.
- **Relevant:** Improve access to rainfall data from remote SAEON sites.
- **Time-bound: Follow the 13-week project schedule from 28 July to 23 October 2026, Tuesday to Friday. Deployment and field validation are still In Progress at this reporting cut-off.**

Project Milestones

Milestone	Description	Timeline
Project Planning and Architecture	Scope, roles, component choices and system architecture	Week 1
Firmware, Cloud and Cellular	Sensor firmware, HiveMQ, Supabase and SIM7000G connectivity	Weeks 2–4
Dashboard, Power and Diagnostics	Live/history views, CSV export, solar integration and station health	Weeks 5–7
Assembly and Deployment Preparation	PCB assembly, enclosure, power audit and solar checks	Weeks 8–9
Field Validation	Deployment, baseline calibration and 48-hour field testing; In Progress	Weeks 9–10; ongoing at cut-off
Documentation and Prototype Review	Slides, rehearsals and prototype documentation sign-off; field acceptance remains open	Weeks 11–13

10. Risks & Constraints

Known Risks

- Missed deadlines; sensor selection and calibration issues.
- Poor communication; cellular or cloud connectivity loss.
- Unequal contribution; outdoor power and weather exposure.

Dependencies

- Cellular coverage and service; HiveMQ Cloud access.
- Team availability; controller, sensors, and prototyping materials.
- Lecturer feedback; SAEON requirements and site access.

Constraints

- Limited time; final deployment and field validation remain In Progress.
- Budget limitations; deployment power and enclosure needs.
- Skill gaps in embedded systems, MQTT, and sensor integration.

11. Review & Improvement

- **Charter Review Frequency:** Every Retrospective.
- **Process Improvements:** Made after each milestone.
- **Feedback:** Lecturer feedback incorporated continuously.

12. Team Commitment

By working on this team, we commit to the principles and agreements outlined in this charter.

Name	Signature / Agreement
Angelo Adams	A.Adams
Redah Gamieldien	R. Gamieldien
Qaasim Isaacs	Q. Isaacs
Ethan Williams	E. Williams
Rameez Karriem	R. Karriem